

COMM144 Generative AI 2nd Workshop

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Exercise 1

Practice with python programming for training a VAE in the python notebook.

Exercise 2

check the [website link](#) on how to derive the final training loss for $D_{KL}(q_{\varphi}(\mathbf{z}|\mathbf{x}_i)||p_{\theta}(\mathbf{z}))$.
That is, to show that

$$\begin{aligned} D_{KL}(\mathcal{N}(\mathbf{z}; \boldsymbol{\mu}^{(i)}, \boldsymbol{\sigma}^{2,(i)} \mathbf{I}) || \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})) \\ = \frac{1}{2} \sum_{j=1}^k \left(1 + \log(\boldsymbol{\sigma}^{(i)}[j])^2 - (\boldsymbol{\mu}^{(i)}[j])^2 - (\boldsymbol{\sigma}^{(i)}[j])^2 \right). \end{aligned}$$

Remarks:

- $\text{tr}(A)$ denotes the trace of a matrix A . Check [website link](#) on how to compute the trace.
- $|\cdot|$ denotes the determinant of a matrix A . Check [website link](#) on how to compute the determinant.